

Finding the Day of the Week in 'C' for a Given Date

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Generally, a novice starts programming with 'C' language—the most trusted language in the embedded field. 'C' language is quite easy to learn, robust and faster than other higher-level languages like Java and gives immense scope and flexibility due to its very basic nature.

Presented here is a utility program written in 'C'. This program is mainly meant for students and hobbyists who are new to 'C' and want to understand its basic concepts.

It often happens that we need to know the day of the week for any past or future date. If you have a wall or table calendar, you can easily find that 4th March, 2012, fell on Sunday. This program exactly does the same and informs what was the day on a given date.

The program is rather simple, and uses the concepts of 'if', 'for' and 'switch' loops and some other simple 'C' functions. The compiler used is Turbo C++ V3.0, which is easily available on the website and can be used by students and professionals alike. There is a 'time.h' header file available

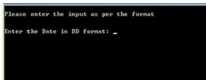


Fig. 1: Screenshot of the program output



Fig. 2: Screenshot of the input date given by the user

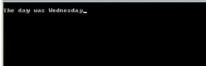


Fig. 3: Screenshot of the final program output

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

on the Internet for 'C' programming, using which you can easily find date and other time related functionalities.



This program has been made without using the time.h header file so that the logic is exploited to the maximum. For details of the time.h header file, you can visit the links given below:

<http://www.cplusplus.com/reference/ctime/>

http://en.wikipedia.org/wiki/C_date_and_time_functions

Program testing

The program (calendar.c) was tested on a Windows XP machine as follows:

1. Install Turbo C++ V3.0 on your system
2. Copy the calendar.c code into your system from EFY DVD of this month's issue
3. Click the TC.exe file in 'bin' folder. Compile the calendar.c code. The output will be as shown in Fig. 1. You should enter the input as per the format given on the screen (refer Fig. 2). The final output should be as shown in Fig. 3.

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CALENDAR.C

```
#include<stdio.h>
#include<conio.h>

int correct_date_calc(int day, int month, int year)
{
    const int ref_year=2012;
    int lp_yr_flg=0;

    if(year<=2012)
    {
        if(((ref_year - year)%4) == 0)
        {
            lp_yr_flg=1;
        }
    }
    else if(year>2012)
    {
        if(((lp_yr_flg==0) && (month==2) && (day>29))
        {
            return 0;
        }
        else if((lp_yr_flg==0) && (month==2) && (day>28))
        {
            return 0;
        }
    }
    else if((month == 3) || (month == 7) || (month == 8) || (month == 10) || (month == 12)) && (day>31 || day<1)
    {
        return 0;
    }
    else if((month == 4) || (month == 6) || (month == 9) || (month == 11)) && (day>30 || day<1)
    {
        return 0;
    }
    return 0;
}
```